

Duality

$$\begin{array}{ll} \text{(P)} & \max \quad c^T x \\ & \text{s.t.} \quad Ax \leq b \\ & \quad \quad x \geq 0 \end{array}$$

y^T

$$y^T Ax \leq y^T b$$

$$c^T x \leq y^T Ax \leq y^T b$$

$$c^T \leq y^T A$$

$$y \geq 0$$

$$(x \geq 0)$$

$$\begin{array}{ll} \text{(D)} & \min \quad b^T y \\ & \text{s.t.} \quad A^T y \geq c \\ & \quad \quad y \geq 0 \end{array}$$

Complementary Slackness (at optimal)

$$c^T X \neq y^T A X \neq y^T b$$


Diagram illustrating the relationship between the three terms: $c^T X$, $y^T A X$, and $y^T b$. Red arrows indicate that $c^T X = y^T A X$ and $y^T A X = y^T b$, which implies $c^T X = y^T b$.

$$c^T X = y^T A X$$

$$0 = \underbrace{(y^T A - c^T)}_z X$$

$$0 = z^T X$$

$$x_j z_j = 0$$

$$y^T A X = y^T b$$
$$y^T \underbrace{(A X - b)}_w = 0$$

$$y^T w = 0$$

$$y_i w_i = 0$$

THE GRAPHIC METHOD

[c] p. 260

The geometric interpretation of linear programming leads to a geometric procedure for solving LP problems with two variables. For illustration, let us consider the problem

$$\begin{array}{ll} \text{maximize} & x_1 + x_2 \\ \text{subject to} & 2x_1 + x_2 \leq 14 \\ & -x_1 + 2x_2 \leq 8 \\ & 2x_1 - x_2 \leq 10 \\ & x_1, x_2 \geq 0. \end{array} \quad (17.4)$$

Its region of feasibility is shown in Figure 17.7, together with the line representing all the points

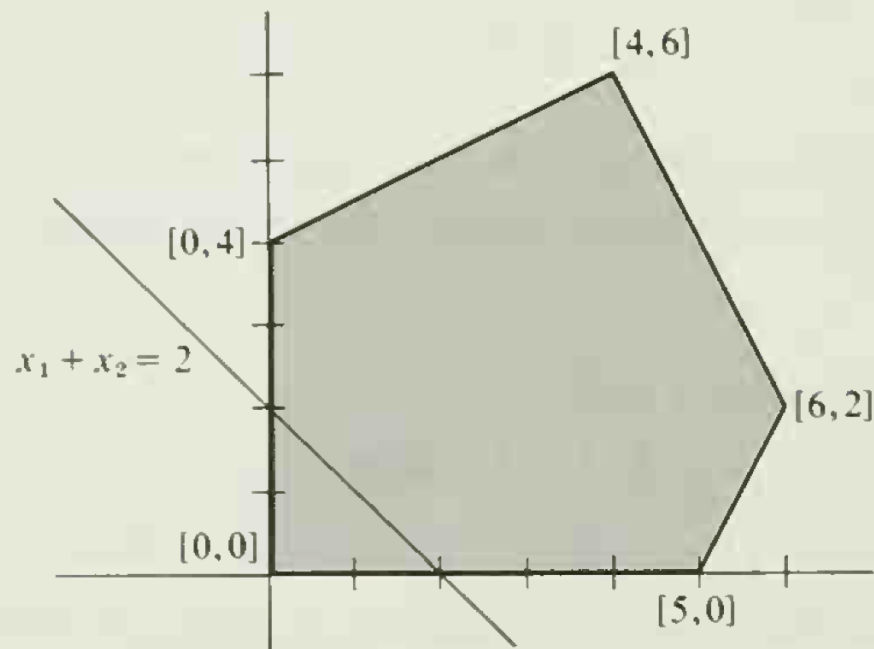
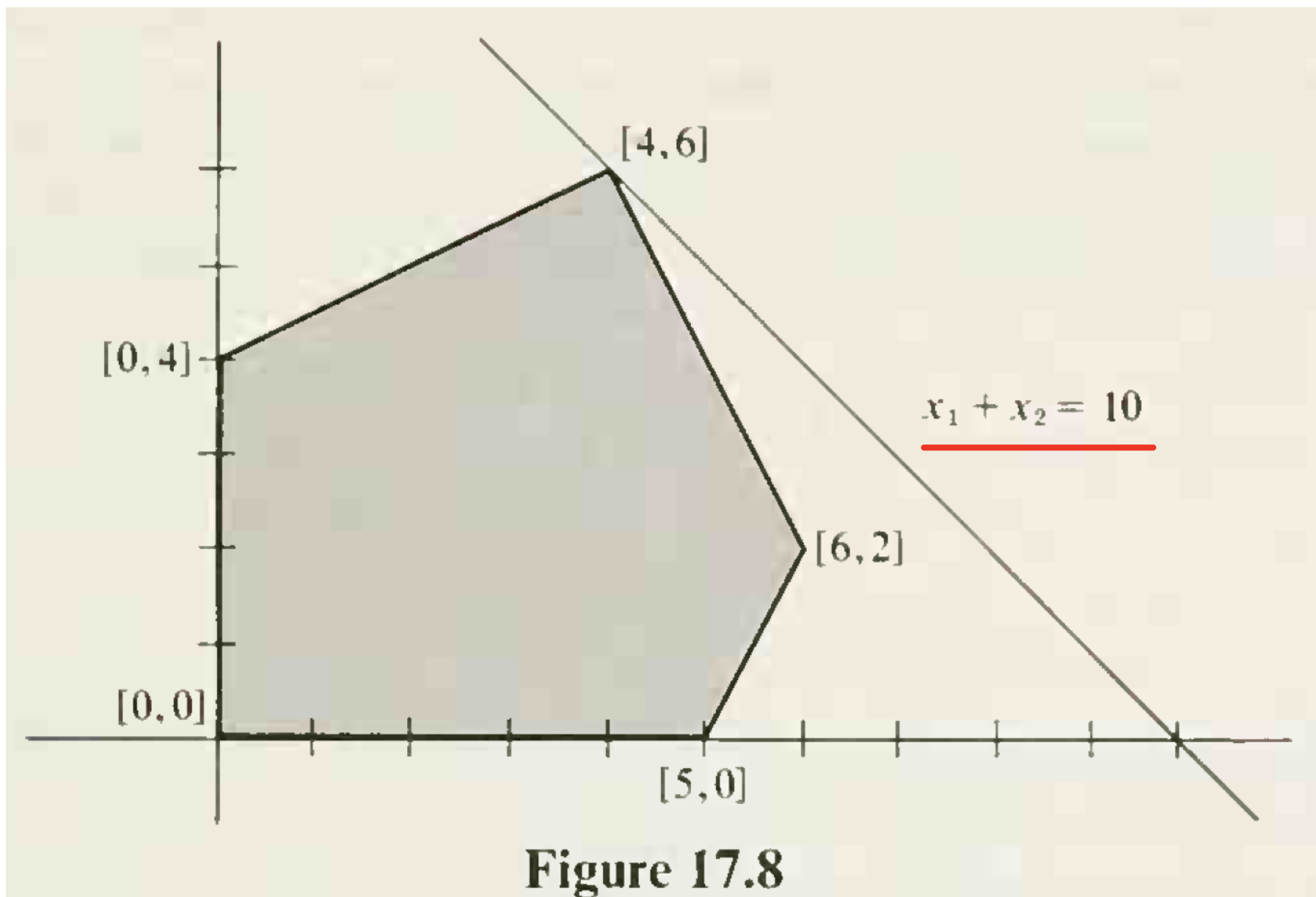


Figure 17.7



$$\max \quad x_1 + x_2$$

$$(P) \quad \text{s.t.} \quad \begin{cases} 2x_1 + x_2 \leq 14 \\ -x_1 + 2x_2 \leq 8 \\ 2x_1 - x_2 \leq 10 \\ x_1, x_2 \geq 0 \end{cases}$$

$$\begin{cases} y_1(2x_1 + x_2) \leq 14y_1 \\ y_2(-x_1 + 2x_2) \leq 8y_2 \\ y_3(2x_1 - x_2) \leq 10y_3 \end{cases}$$

$$(2y_1 - y_2 + 2y_3)x_1 + (y_1 + 2y_2 - y_3)x_2 \leq 14y_1 + 8y_2 + 10y_3$$

$$\max \quad x_1 + x_2$$

$$(P) \quad \text{s.t.} \quad \begin{cases} 2x_1 + x_2 \leq 14 \\ -x_1 + 2x_2 \leq 8 \\ 2x_1 - x_2 \leq 10 \\ x_1, x_2 \geq 0 \end{cases}$$

$$(D) \quad \min \quad 14y_1 + 8y_2 + 10y_3$$

$$\begin{cases} 2y_1 - y_2 + 2y_3 \geq 1 \\ y_1 + 2y_2 - y_3 \geq 1 \\ y_1, y_2, y_3 \geq 0 \end{cases}$$

$$x_1 = 4, x_2 = 6$$

$$\max \quad x_1 + x_2$$

$$(P) \quad \text{s.t.} \quad \begin{cases} 2x_1 + x_2 \leq 14 \\ -x_1 + 2x_2 \leq 8 \\ 2x_1 - x_2 \leq 10 \\ x_1, x_2 \geq 0 \end{cases}$$

$$(D) \quad \min \quad 14y_1 + 8y_2 + 10y_3$$
$$\begin{cases} 2y_1 - y_2 + 2y_3 \geq 1 \\ y_1 + 2y_2 - y_3 \geq 1 \\ y_1, y_2, y_3 \geq 0 \end{cases}$$

$$\begin{array}{ll}
 \max & x_1 + x_2 = 10 \\
 (P) \quad \text{s.t.} & \begin{cases} 2x_1 + x_2 \leq 14 \leftarrow w_1 = 0 \\ -x_1 + 2x_2 \leq 8 \leftarrow w_2 = 0 \\ 2x_1 - x_2 \leq 10 \leftarrow w_3 > 0 \Rightarrow \underline{y_3 = 0} \\ x_1, x_2 \geq 0 \end{cases}
 \end{array}$$

$x_1 = 4, x_2 = 6$
 $z_1 = 0, z_2 = 0$

$$\begin{array}{ll}
 (D) \quad \min & 14y_1 + 8y_2 + 10y_3 = 10 \\
 & \begin{cases} 2y_1 - y_2 + 2y_3 \geq 1 \leftarrow z_1 = 0 \\ y_1 + 2y_2 - y_3 \geq 1 \leftarrow z_2 = 0 \\ y_1, y_2, y_3 \geq 0 \end{cases}
 \end{array}$$

$\underline{y_1 = \frac{3}{5}, y_2 = \frac{1}{5}}$

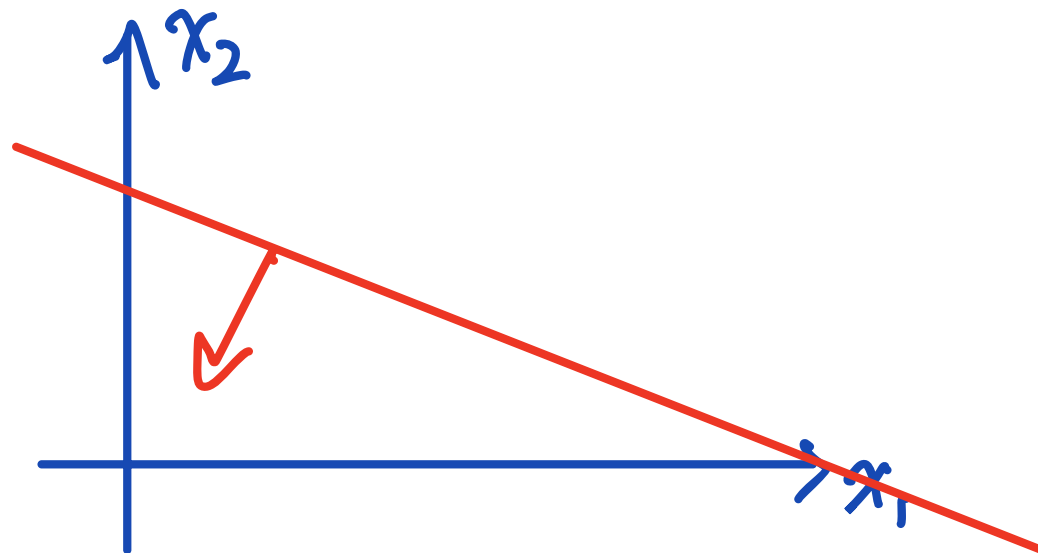
Interpretation of

$$\underline{(2y_1 - y_2 + 2y_3)x_1 + (y_1 + 2y_2 - y_3)x_2 \leq}$$

$$\underline{14y_1 + 8y_2 + 10y_3}$$

$$p(y_1, y_2, y_3)x_1 + q(y_1, y_2, y_3)x_2 \leq r(y_1, y_2, y_3)$$

half space



$$y_3 = 0 \Rightarrow$$

$$(2y_1 - y_2)x_1 + (y_1 + 2y_2)x_2 \leq 14y_1 + 8y_2$$

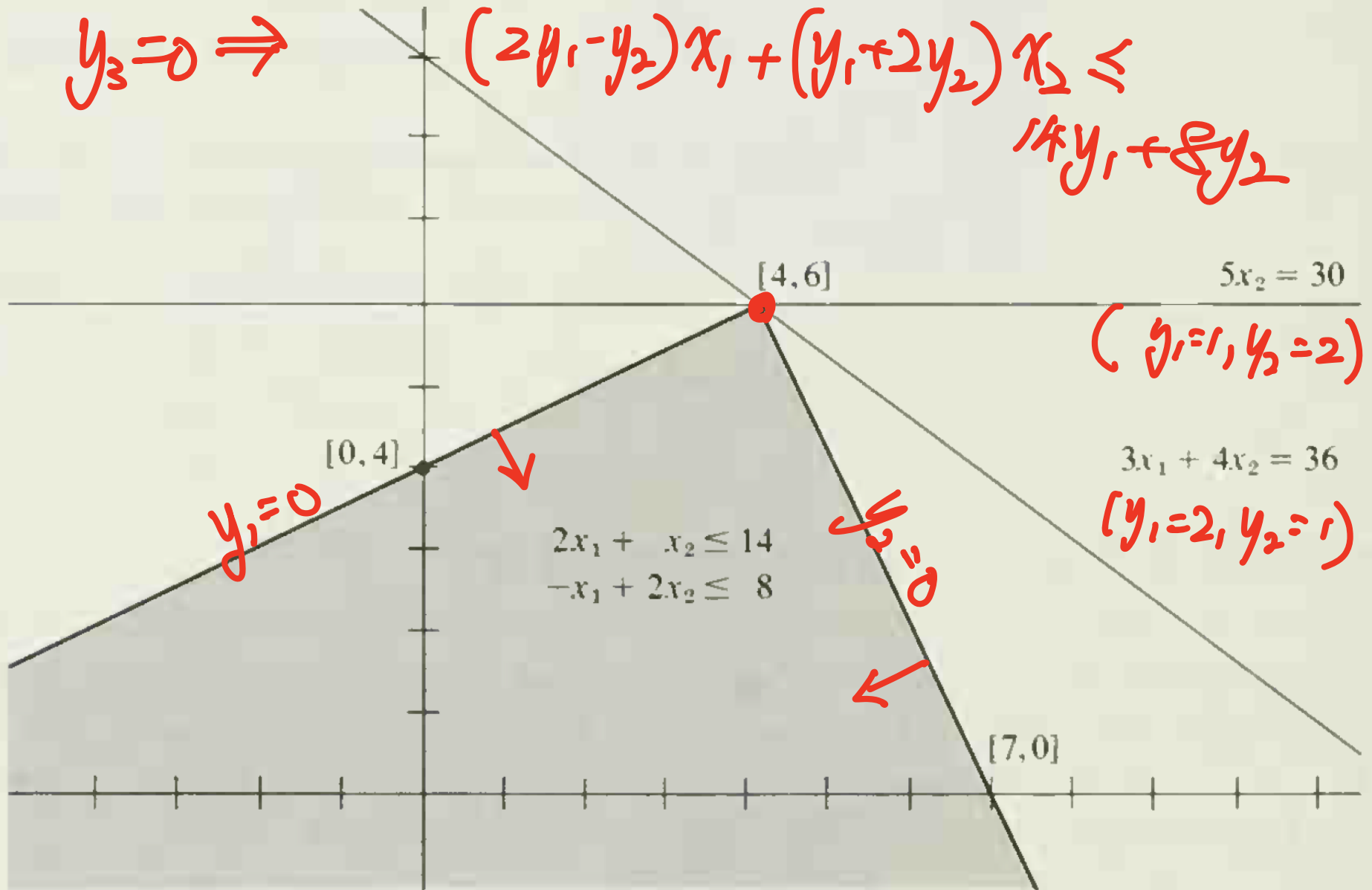


Figure 17.9

Sensitivity Analysis (what if c, b change)

$$\begin{aligned} (\text{P}) \quad & \max \quad J = c^T x \\ & \text{s.t.} \quad Ax \leq b \\ & \quad \quad x \geq 0 \end{aligned}$$

$$\begin{aligned} (\text{D}) \quad & \min \quad \xi = b^T y \\ & \text{s.t.} \quad A^T y \geq c \\ & \quad \quad y \geq 0 \end{aligned}$$

Sensitivity Analysis (what if c, b change)

$$(P) \max \quad \hat{J} = c^T x \\ \text{s.t.} \quad Ax \leq b \\ x \geq 0$$

$$(D) \min \quad \hat{J} = b^T y \\ \text{s.t.} \quad A^T y \geq c \\ y \geq 0$$

$$\hat{J}(x_B, x_N) = c_B^T \bar{B}^{-1} b - ((\bar{B}^{-1} N)^T c_B - c_N)^T x_N$$

$$x_B = \bar{B}^{-1} b - (\bar{B}^{-1} N) x_N$$

at opt.

$$x_N^* = 0, \quad x_B^* = \bar{B}^{-1} b \geq 0$$

$$(\bar{B}^{-1} N)^T c_B - c_N \geq 0$$

$$\hat{J}^* = c_B^T x_B^* = c^T x^*$$

Sensitivity Analysis (what if c, b change)

$$(P) \max \quad \bar{z} = c^T x \\ \text{s.t.} \quad Ax \leq b \\ x \geq 0$$

$$(D) \min \quad \bar{z} = b^T y \\ \text{s.t.} \quad A^T y \geq c \\ y \geq 0$$

$$- \bar{z}(z_B, z_N) = -c_B^T \bar{B}^{-1} b - (\bar{B}^{-1} b)^T z_B$$

$$z_N = ((\bar{B}^{-1} N)^T c_B - \hat{c}_N) + (\bar{B}^{-1} N)^T z_B$$

at opt.

$$z_B^* = 0, \quad z_N^* = (\bar{B}^{-1} N)^T c_B - \hat{c}_N \geq 0$$

$$\bar{B}^{-1} b \geq 0$$

$$\bar{z}^* = c_B^T \bar{B}^{-1} b = b^T y^* \quad (y^* = (\bar{B}^{-1})^T \hat{c}_B)$$

Sensitivity Analysis (what if c, b change)

$$\begin{aligned} (\text{P}) \quad & \max \quad J = c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

$$\begin{aligned} (\text{D}) \quad & \min \quad \tilde{J} = b^T y \\ \text{s.t.} \quad & A^T y \geq c \\ & y \geq 0 \end{aligned}$$

at Opt. $J^* = c^T x^* = b^T y^* = \tilde{J}^*$

if only c changes

$$\Delta J^* = (\Delta c)^T x^*$$

Sensitivity Analysis (what if c, b change)

$$\begin{aligned} (\text{P}) \quad & \max \quad J = c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

$$\begin{aligned} (\text{D}) \quad & \min \quad \xi = b^T y \\ \text{s.t.} \quad & A^T y \geq c \\ & y \geq 0 \end{aligned}$$

at Opt. $J^* = c^T x^* = b^T y^* = \xi^*$

if only b changes

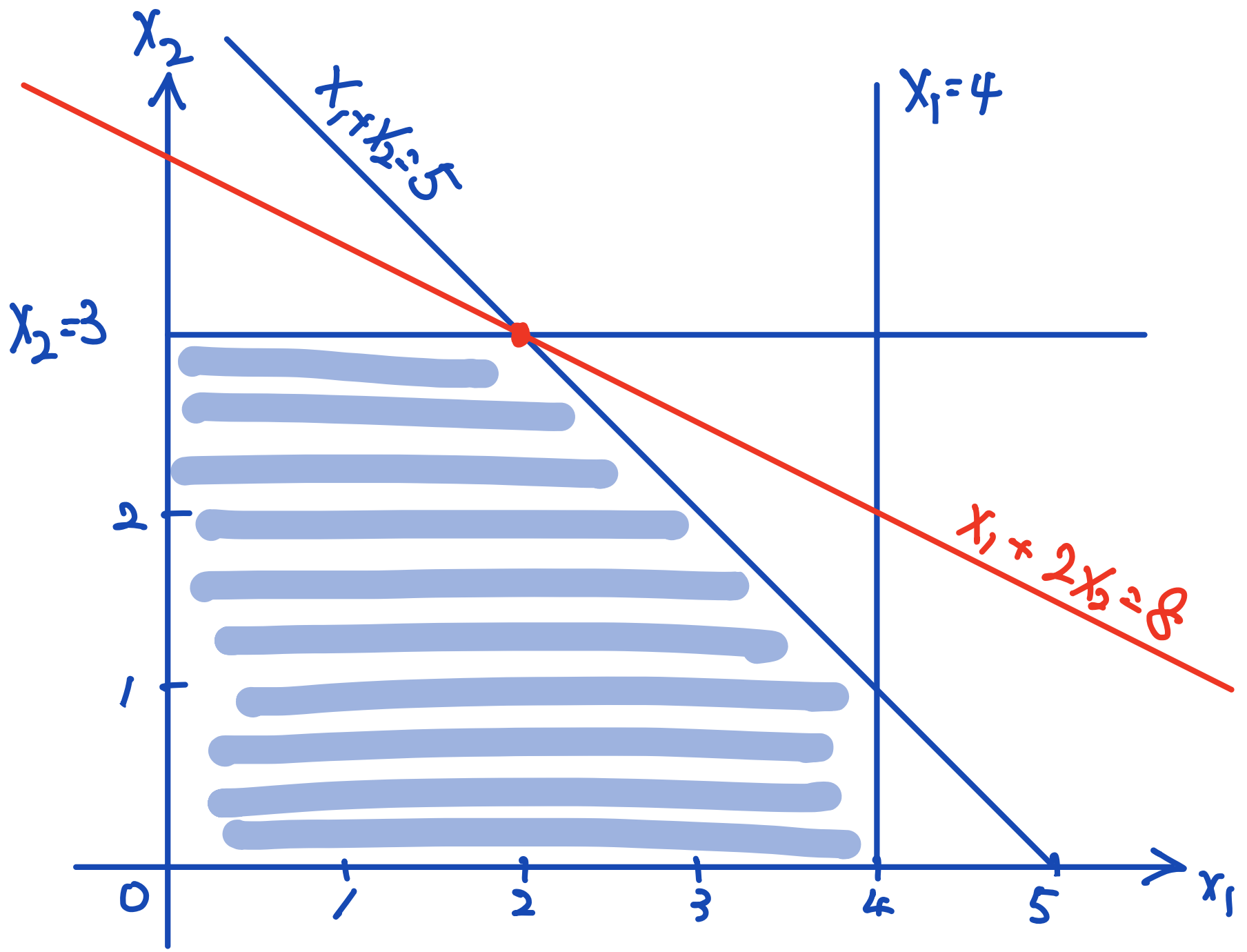
$$\Delta J^* (= \Delta \xi^*) = (\Delta b)^T y^*$$

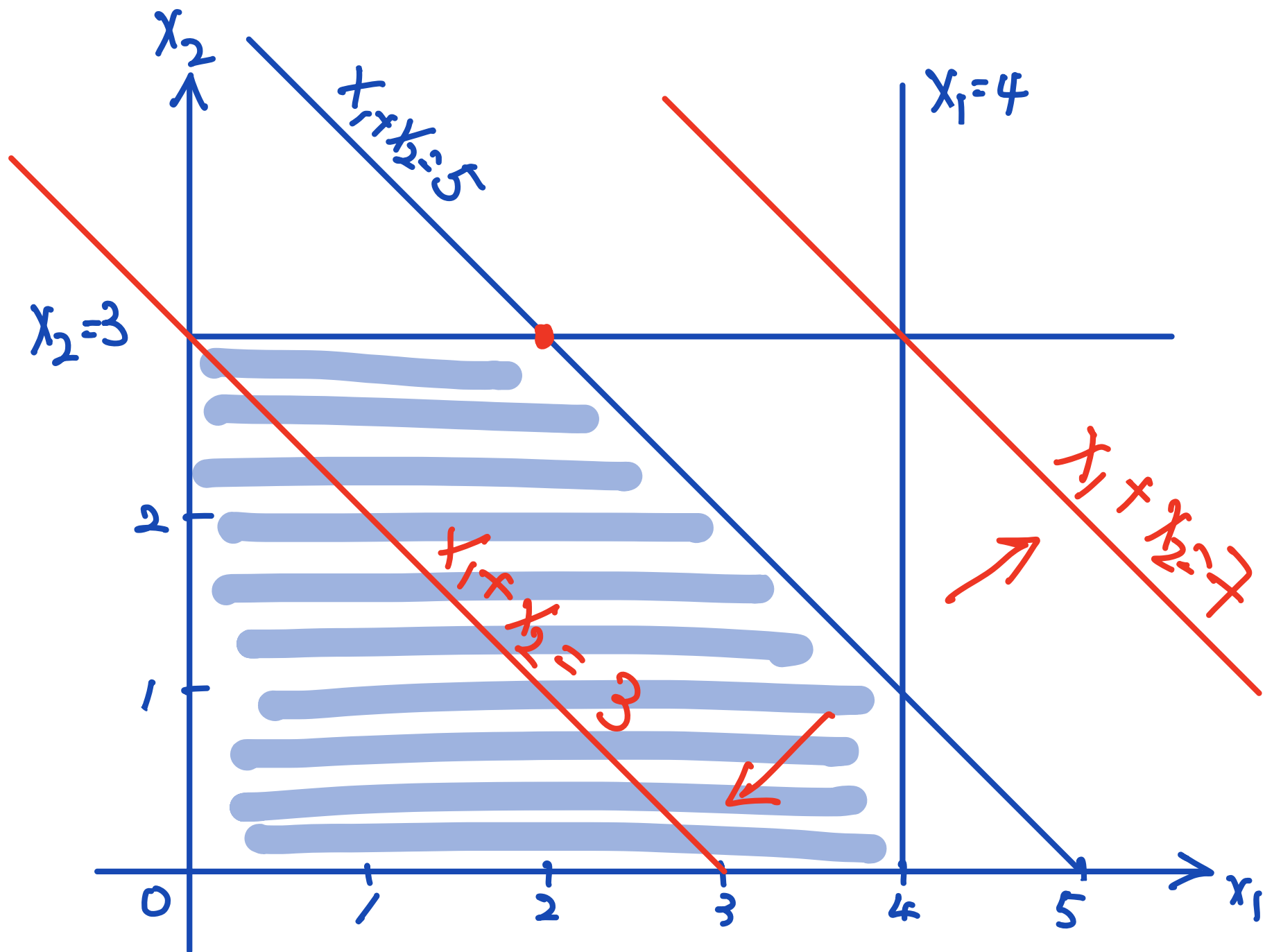
y^* : shadow prices

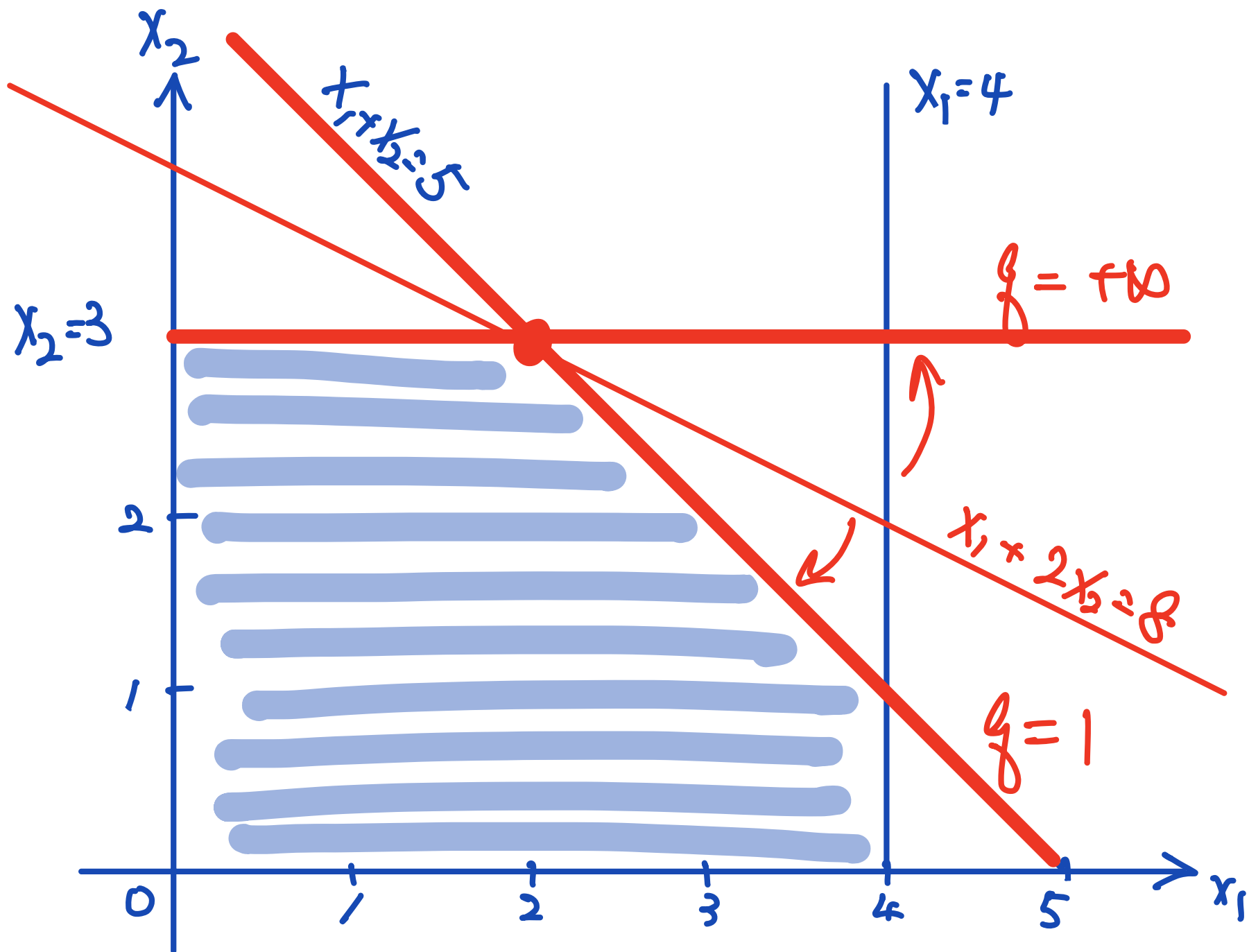
4. Consider the following linear programming problem:

$$\begin{array}{ll}\text{maximize} & x_1 + 2x_2 \\ \text{subject to} & x_1 + x_2 \leq 5 \\ & x_1 \leq 4 \\ & x_2 \leq 3 \\ & x_1, x_2 \geq 0\end{array}$$

- (a) Solve the above problem and give also the dictionary at the optimal point.
- (b) Suppose the constraint $x_1 + x_2 \leq 5$ is changed to $x_1 + x_2 \leq p$. Find the range of p such that the dictionary you have found in (a) remains optimal.
- (c) Suppose the objective function $x_1 + 2x_2$ is changed to $x_1 + qx_2$. Find the range of q such that the dictionary you have found in (a) remains optimal.







Sensitivity Analysis (what if ~~c, b~~ A changes)

$$J^* = C_B^T B^{-1} b = J^* \\ (= C^T X^*) \quad (= b^T y^*)$$

$$\Delta J^* = C_B^T (\Delta B^{-1}) b$$

$$\underline{\Delta(B^{-1}) = -B^{-1}(\Delta B)B^{-1}}$$

$$\Delta J^* = -C_B^T B^{-1}(\Delta B)B^{-1} b$$

Sensitivity Analysis (what if ~~c, b~~ _A changes)

$$J^* = C_B^T B^{-1} b = J^* \\ (= C^T x^*) \quad (= b^T y^*)$$

$$\Delta J^* = - \underbrace{C_B^T}_{(y^*)^T} B^T (\Delta B) \underbrace{B^{-1} b}_{x_B^*}$$

$$= - (y^*)^T (\Delta B) x_B^* = - \langle (\Delta B) x_B^*, y^* \rangle$$

Ex (Mid term #4)

$$\max \quad x_1 + 2x_2$$

$$\text{s.t.} \quad x_1 + x_2 \leq 5$$

$$x_1 \leq 4$$

$$x_2 \leq 3$$

$$x_1, x_2 \geq 0$$

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 3 \end{bmatrix}$$

$\underbrace{\hspace{1.5cm}}_A \quad \underbrace{\hspace{1.5cm}}_N$

Ex (Mid term #4)

$$\max \quad x_1 + 2x_2$$

$$\text{s.t.} \quad x_1 + x_2 \leq 5$$

$$x_1 \leq 4$$

$$x_2 \leq 3$$

$$x_1, x_2 \geq 0$$

at Opt:

$$x_1^* = 2, x_2^* = 3$$

$$w_1^* = 0$$

$$w_2^* > 0$$

$$w_3^* = 0$$

$$\left[\underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}}_A \quad \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_N \right] \begin{bmatrix} x_1 \\ x_2 \\ w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \Theta = \{1, 2, 4\}$$

1 2 4

$$x_B^* = B^{-1}b$$

$$= \begin{pmatrix} x_1 \\ x_2 \\ w_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 5 \\ 4 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 5 \\ 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}$$

$\leftarrow x_1^*$
 $\leftarrow x_2^*$
 $\leftarrow w_2^*$

$$B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \Theta = \{1, 2, 4\}$$

1 2 4

$$y^* = (B^{-1})^T C_\Theta = \left[\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}^{-1} \right]^T \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{matrix} \leftarrow y_1^* \\ \leftarrow y_2^* \\ \leftarrow y_3^* \end{matrix}$$

$$J = C^T B^{-1} b$$

$$\Delta J = C^T \Delta(B^{-1}) b$$

$$= -C^T B^{-1} (\Delta B) B^{-1} b$$

$$= -((B^{-1})^T C)^T \Delta B (B^{-1} b)$$

$$= -(y^*)^T \Delta B x_B^*$$

$$= - \begin{pmatrix} 1 & 0 & 1 \end{pmatrix} (\Delta B) \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}$$

$$\Delta f = - \begin{pmatrix} 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \Delta a_{11} & \Delta a_{12} & 0 \\ \Delta a_{21} & \Delta a_{22} & 0 \\ \Delta a_{31} & \Delta a_{23} & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}$$

$$= - \begin{pmatrix} \Delta a_{11} + \Delta a_{31} & \Delta a_{12} + \Delta a_{23} & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}$$

$$= - (2\Delta a_{11} + 2\Delta a_{31} + 3\Delta a_{12} + 3\Delta a_{23})$$

$$X_B^* = B^{-1}b$$

$$\Delta X_B^* = \Delta(B^{-1})b$$

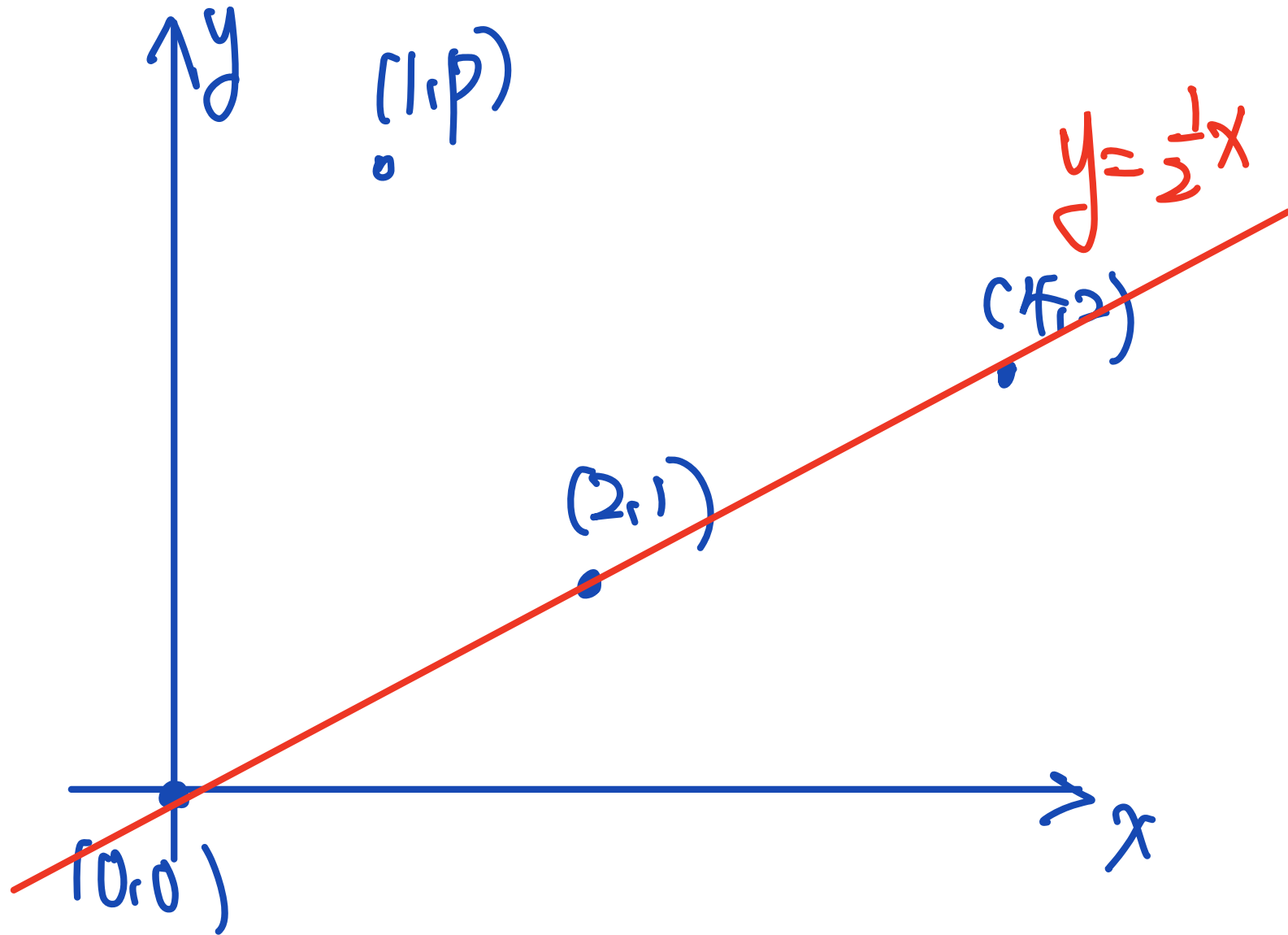
$$= -B^{-1}(\Delta B)B^{-1}b$$

$$y^* = (B^{-1})^T c_B$$

$$\Delta y^* = -\left(B^{-1}(\Delta B)B^{-1}\right)^T c_B$$

$$= -\left(B^{-1}\right)^T (\Delta B)^T \left(B^{-1}\right)^T c_B$$

HW 9 #2



$$y = ax + b \quad (a = \frac{1}{2}, b = 0)$$

$$(P) \quad \min \quad t_1 + t_2 + t_3 + t_4$$

$$\text{s.t.} \quad |0 - (a(0) + b)| \leq t_1$$

$$|1 - (a(2) + b)| \leq t_2$$

$$|2 - (a(4) + b)| \leq t_3$$

$$|p - (a(1) + b)| \leq t_4$$

$$y = ax + b \quad (a = \frac{1}{2}, b = 0)$$

$$(P) \quad \min \quad t_1 + t_2 + t_3 + t_4$$

$$\text{s.t.} \quad -t_1 \leq -b \leq t_1$$

$$-t_2 \leq 1 - 2a - b \leq t_2$$

$$-t_3 \leq 2 - 4a - b \leq t_3$$

$$-t_4 \leq 1 - a - b \leq t_4$$

Proposed solution

$a = \frac{1}{2}, b = 0$

$\Rightarrow$

$$-t_1 \leq 0 \leq t_1 \Rightarrow t_1 = 0$$

$$-t_2 \leq 0 \leq t_2 \Rightarrow t_2 = 0$$

$$-t_3 \leq 0 \leq t_3 \Rightarrow t_3 = 0$$

$$-t_4 \leq p - \frac{1}{2} \leq t_4$$

$$\Rightarrow \underline{t_4 = |p - \frac{1}{2}|}$$

$$\underline{J^* = |p - \frac{1}{2}|}$$

$$u_1(t_1 + b \geq 0)$$

$$u_2(t_1 - b \geq 0)$$

$$u_3(t_2 + 2a + b \geq 1)$$

$$u_4(t_2 - 2a - b \geq -1)$$

$$u_5(t_3 + 4a + b \geq 2)$$

$$u_6(t_3 - 4a - b \geq -2)$$

$$u_7(t_4 + a + b \geq p)$$

$$u_8(t_4 - a - b \geq -p)$$

$$u_i \geq 0$$

$$\max (u_3 - u_4) + 2(u_5 - u_6) + 1/4(u_7 - u_8)$$

s.t.

$$u_1 + u_2 = 1,$$

$$u_3 + u_4 = 1$$

$$u_5 + u_6 = 1$$

$$u_7 + u_8 = 1$$

$$2(u_3 - u_4) + 1/4(u_5 - u_6) + (u_7 - u_8) = 0$$

$$\underbrace{(u_1 - u_2)}_{v_1} + \underbrace{(u_3 - u_4)}_{v_2} + \underbrace{(u_5 - u_6)}_{v_3} + \underbrace{(u_7 - u_8)}_{v_4} = 0$$

$$\begin{cases}
 u_1 (t_1 + \cancel{b} \geq \cancel{0}) \\
 u_2 (t_1 - \cancel{b} \geq \cancel{0}) \\
 u_3 (t_2 + \cancel{2a+b} \geq \cancel{1}) \\
 u_4 (t_2 - \cancel{2a-b} \geq \cancel{-1}) \\
 u_5 (t_3 + \cancel{4a+b} \geq \cancel{2}) \\
 u_6 (t_3 - \cancel{4a-b} \geq \cancel{-2}) \\
 u_7 (t_4 + a + b \geq p) \\
 u_8 (t_4 - a - b \geq -p)
 \end{cases}$$

$$a = \frac{1}{2}, b = 0$$

$$u_i \geq 0$$

(Cannot say anything about $u_1, u_2, u_3, u_4, u_5, u_6$, other than $u_i \geq 0$)

For u_7, u_8 :

$$u_7 \left(t_4 + \frac{1}{2} \geq p \right)$$

$$u_8 \left(t_4 - \frac{1}{2} \geq -p \right)$$

If $p > \frac{1}{2}$, then $t_4 = p - \frac{1}{2} \Rightarrow u_8 = 0, u_7 = 1$

If $p < \frac{1}{2}$, then $t_4 = \frac{1}{2} - p \Rightarrow u_7 = 0, u_8 = 1$

For $p \geq \frac{1}{2}$,

$$\max \quad v_2 + 2v_3 + p$$

$$\text{s.t.} \quad 2v_2 + 4v_3 + 1 = 0$$

$$v_1 + v_2 + v_3 + 1 = 0$$

$$\max \quad p - \frac{1}{2}$$

$$\text{s.t.} \quad v_2 + 2v_3 = -\frac{1}{2}$$

$$v_1 + v_2 + v_3 = -1$$

Make Sure
this is
feasible

For $p < \frac{1}{2}$,

$$\max \quad v_2 + 2v_3 - p$$

$$\text{s.t.} \quad 2v_2 + 4v_3 - 1 = 0$$

$$v_1 + v_2 + v_3 - 1 = 0$$

$$\max \quad \frac{1}{2} - p$$

$$\text{s.t.} \quad v_2 + 2v_3 = \frac{1}{2}$$

$$v_1 + v_2 + v_3 = 1$$

Make Sure
this is
feasible